TOM MBOYA UNIVERSITY

KNOWLEDGE FOR SUSTAINABLE INNOVATION ENTERPRISE

2025/2026 ACADEMIC YEAR UNIVERSITY EXAMINATION

FOURTH YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION(SCIENCE)WITH IT

BACHELOR OF EDUCATION(ARTS) WITH IT
BACHELOR OF SCIENCE WITH IT

BACHELOR OF SCIENCE ACTUARIAL SCIENCE WITH IT

MMA 405: PARTIAL DIFFERENTIAL EQUATIONS

DATE: 16/12/2025 TIME :3 HOURS

INSTRUCTIONS

1. Do not write anything on this question paper.
2. Answers question ONE and any other TWO questions.

EXAMINATION CHEATING IS ACRIME

QUESTION ONE (30 MARKS)

a) Define the following terms

(i) Partial Differential Equations

(ii) Fourier Transform

(iii) Delta function (6marks)

b) Solve the equation $r = 6x$ (6marks)

c) Find the Fourier Sine Transform of $\frac{1}{x}$ (6marks)

d) Derive the Green's function for the operator $\frac{d^2}{dx^2}$ with the boundary conditions $y(0) = 0 \; and \; y(1) = 0$ (6marks)

e) Show that the equation $(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$ is exact and hence solve it. (6marks)

QUESTION TWO (30 MARKS)

a) Find the adjoint and hence the inverse of the matrix A below

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ -1 & 3 & 0 \end{pmatrix}$$ (12marks

b) Derive the Green's function for the operator $\frac{d^2}{dx^2}$ with the boundary conditions $y(0) = 0 \; and \; y(1) = 0$ (8marks)

QUESTION THREE (20 MARKS)

a)For the equation $(8y - x^2y)dy + (x - xy^2)dx = 0$

i) Show that it is exact

ii) Find its solution (10 marks)

b) Given $A = 2i + 2j - k$ and $b = 6i - 3j + 2k$

i) Find the angle between A and B

ii) Find $A \times B$ (10 marks)

QUESTION FOUR (20 MARKS)

a) Find the Fourier Sine Transform of $2e^{-3x} + 3e^{-2x}$ (10 marks)

b) Find the Fourier Cosine Transform of $f(x) = \frac{1}{1+x^2}$ (10marks)

QUESTION FIVE (20 MARKS)

a) Given $\emptyset\,(x,y,z) = 3x^2y - y^3z^2$, find $\nabla\emptyset$(or grad $\emptyset$) at the point $(1,-2,-1)$ (7 marks)

b) Given $A = x^2zi - 2y^3z^2j + xy^2zk$, find $\nabla \cdot A$(or divA) at the point $(1,-1,1)$ (6mks)

c) If $A = xz^3i - 2x^2yzj + 2yz^4k$, find $\nabla \times A$ (or curl curl A) (7 marks)